

Towards Explainable Fusion and Balanced Learning in Multimodal Sentiment Analysis

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Abstract

Multimodal Sentiment Analysis (MSA) faces two critical challenges: the lack of interpretability in the decision logic of multimodal fusion and modality imbalance caused by disparities in inter-modal information density. To address these issues, we propose KAN-MCP, a novel framework that integrates the interpretability of Kolmogorov-Arnold Networks (KAN) with the robustness of the Multimodal Clean Pareto (MCPareto) framework. First, KAN leverages its univariate function decomposition to achieve transparent analysis of cross-modal interactions. This structural design allows direct inspection of feature transformations without relying on external interpretation tools, thereby ensuring both high expressiveness and interpretability. Second, the proposed MCPareto enhances robustness by addressing modality imbalance and noise interference. Specifically, we introduce the Dimensionality Reduction and Denoising Modal Information Bottleneck (DRD-MIB) method, which jointly denoises and reduces feature dimensionality. This approach provides KAN with discriminative low-dimensional inputs to reduce the modeling complexity of KAN while preserving critical sentiment-related information. Furthermore, MCPareto dynamically balances gradient contributions across modalities using the purified features output by DRD-MIB, ensuring lossless transmission of auxiliary signals and effectively alleviating modality imbalance. This synergy of interpretability and robustness not only achieves superior performance on benchmark datasets such as CMU-MOSI, CMU-MOSEI, and CH-SIMS v2 but also offers an intuitive visualization interface through KAN's interpretable architecture. Our code is released on <https://github.com/LuoMSen/KAN-MCP>.

CCS Concepts

- **Information systems** → **Multimedia information systems**;
- **Computing methodologies** → **Machine learning**; Natural language processing; Supervised learning by regression.

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MM '25, Dublin, Ireland

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ACM ISBN 979-8-4007-2035-2/2025/10
<https://doi.org/10.1145/3746027.3755591>

Keywords

Multimodal Sentiment Analysis, Kolmogorov-Arnold Networks, Information Bottleneck, Imbalanced Multimodal Learning

ACM Reference Format:

Miaosen Luo, Yuncheng Jiang, and Sijie Mai. 2025. Towards Explainable Fusion and Balanced Learning in Multimodal Sentiment Analysis. In *Proceedings of the 33rd ACM International Conference on Multimedia (MM '25)*, October 27–31, 2025, Dublin, Ireland. ACM, New York, NY, USA, 10 pages. <https://doi.org/10.1145/3746027.3755591>

1 Introduction

Multimodal Sentiment Analysis (MSA) aims to achieve fine-grained modeling and analysis of human sentiments by integrating multi-source information from text, speech, visual and other modalities [62]. Compared to traditional methods relying on single modalities (e.g., text or audio), MSA captures richer sentimental cues through cross-modal complementarity, such as combining sarcasm detection in speech intonation with subtle facial micro-expression variations [18]. This technology has shown significant value in affective computing, mental health assessment, and intelligent interaction [16].

However, while multimodal fusion significantly enhances the performance of the model, the increasing complexity of cross-modal interactions obscures the internal feature fusion logic, resulting in opaque decision-making processes that severely hinder model trustworthiness and optimization [1]. To improve the interpretability of deep learning models, researchers have proposed two categories of solutions from the perspective of explainable AI [39]. Traditional post-hoc interpretability methods [29], such as LIME [7] and SHAP [28], indirectly infer model logic through local approximations or feature importance analysis. However, their explanations may suffer from approximation errors and do not directly reflect the actual decision-making process of the model. In contrast, ante-hoc interpretable methods, such as linear models and decision trees, offer structural transparency but lack expressive power for modeling complex nonlinear relationships.

To address these limitations, Kolmogorov-Arnold Networks (KAN) [23], based on the Kolmogorov-Arnold representation theorem [40], exhibit unique advantages. The Kolmogorov-Arnold theorem states that any continuous multivariate function can be represented as a finite composition of univariate continuous functions. KAN implements this theory by placing activation functions on the edges (connections) of the network rather than on the nodes. This innovative architecture ensures that each layer comprises univariate functions directly corresponding to input feature transformations, enabling model behavior to be understood by analyzing these univariate functions without relying on external interpretation tools. KAN

achieves both high expressiveness (approximating arbitrary continuous functions) and interpretability through structural design, overcoming the performance limitations of traditional ante-hoc methods in complex tasks. Furthermore, KAN's visualization capability enhances its interpretability. As shown in Figure 1, by setting the transparency of the activation functions and node connections proportional to their functional magnitudes, users can intuitively identify which data points contribute significantly to the output of the model, thereby focusing on critical features and relationships.

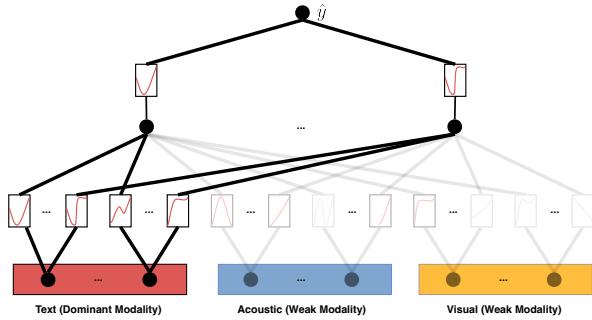


Figure 1: KAN visualization of the text modality dominance effect for MSA, where the transparency of neuron nodes and connection weights gradually becomes less opaque as the feature contribution levels increase.

Despite these advantages, directly applying KAN to MSA remains challenging. When integrating high-dimensional feature vectors from multiple modalities, it is difficult and time-consuming for KAN to analyze the effect of each feature. Moreover, when applying KAN to MSA, we observed that the text modality exhibits significantly higher information density compared to noisy audio and visual modalities [30, 31], causing models to over-rely on text while neglecting contributions from other modalities. To mitigate this, Wei et al. [47] analyzed Pareto integration in multimodal scenarios and proposed the multimodal Pareto algorithm (MMPareto). This algorithm aligns gradient directions across all learning objectives to improve generalization and provides harmless unimodal auxiliary signals, alleviating the modality imbalance. However, the original MMPareto algorithm does not account for unimodal noise interference, leading to performance degradation when non-dominant modalities (e.g., audio or visual) contain significant noise.

Our solution, the Multimodal Clean Pareto (MCPareto) framework, introduces hierarchical optimization. First, in the feature purification layer, we design the Dimensionality Reduction and Denoising Modal Information Bottleneck (DRD-MIB) method based on the Information Bottleneck (IB) principle [3, 32, 42, 50]. DRD-MIB maximizes mutual information between features and sentiment labels while filtering irrelevant noise during feature dimensionality reduction. This process not only achieves feature denoising but also adapts to KAN's univariate function representation capability by reducing feature complexity, ensuring that the model retains minimally sufficient features for effective learning. Subsequently, in the gradient coordination layer, MCPareto leverages the MultiModal Pareto algorithm (MMPareto) [47] on the purified unimodal features generated by DRD-MIB. By implementing dual optimization

mechanisms—gradient direction alignment and adaptive magnitude modulation—MCPareto reconstructs the multimodal Pareto front. This ensures lossless transmission of unimodal auxiliary signals and significantly enhances generalization robustness through cross-modal gradient field coordination.

Building on these innovations, we propose KAN-MCP, a framework that synergizes KAN's interpretability with MCPareto's robustness. Its core contributions address the aforementioned challenges through three key aspects:

- **Interpretable Fusion Mechanism:** We introduce KAN to the MSA field for the first time, enabling transparent cross-modal interaction analysis through univariate function decomposition and visual pathways.
- **Noise Robustness Enhancement:** The proposed DRD-MIB method jointly denoises and reduces feature dimensionality, providing KAN with discriminative low-dimensional inputs and assisting the alleviation of modality imbalance issue.
- **Modality-Balanced Learning:** The MCPareto strategy dynamically balances gradient contributions using purified features from DRD-MIB, effectively alleviating the modality imbalance based on more discriminative unimodal features.

Experiments demonstrate that our method significantly outperforms existing models on the CMU-MOSI, CMU-MOSEI, and CH-SIMS v2 benchmarks. Additionally, KAN's visualization interface intuitively reveals decision logic, offering theoretical tools and practical paradigms for trustworthy deployment of multimodal models.

2 Related Work

2.1 MSA Fusion Methods

To comprehensively capture the complexity of human emotions, multimodal information fusion plays a pivotal role in Multimodal Sentiment Analysis (MSA). By integrating features or decisions from diverse modalities such as audio, visual, and textual data, multimodal fusion methods significantly enhance the accuracy of emotion polarity inference. In recent years, researchers have proposed tensor-based fusion [12, 35], attention-based fusion [14, 20, 24], translation-based fusion [25, 27, 61], and context-aware fusion [26, 45] to improve the efficiency of data fusion.

Despite the notable improvements in emotion recognition accuracy and robustness, these fusion methods introduce new challenges due to their inherent complexity. Intricate fusion mechanisms render the model's decision-making process difficult to interpret intuitively, reducing transparency in real-world applications and complicating further optimization. Thus, balancing model complexity with interpretability while pursuing higher performance has emerged as a critical challenge in MSA.

2.2 Interpretability Methods in MSA

Traditional explainable Artificial Intelligence (xAI) fall into two categories [39]: ante-hoc interpretability, which employs structured designs (e.g., feature importance evaluation [51], rule-based systems [2], attention mechanisms [5]) to build transparent and interpretable models; and post-hoc interpretability, which reverse-engineers trained models via gradient analysis [60], perturbation testing, or visualization techniques [46].

In the MSA domain, Li et al. [19] proposed a quantum-inspired multimodal fusion (QMF) method that models intra-modal feature correlations through quantum superposition, fuses cross-modal decisions via quantum entanglement, and disentangles single-/bi-modal contributions using reduced density matrices. While this approach provides explicit explanations for modal interactions, its quantum state abstraction lacks intuitive mapping to practical features. Wu et al. [51] introduced Interpretable Multimodal Capsule Fusion (IMCF), which explicitly quantifies modal importance via routing coefficients but fails to reveal cross-layer feature transformation logic. Tsai et al. [44] developed a multimodal routing method that achieves local explanations through dynamic weight allocation, yet its interpretability remains confined to numerical weight levels. Khalane et al. [17] applied Gradient SHAP [28] to quantify feature contributions, but local gradients cannot capture the global impact of complex multimodal interactions. Nie et al. [36] utilized integrated gradients to visualize input feature contributions, yet their approach inadequately explains internal decision paths and cross-modal synergies.

This work is the first to introduce KAN into MSA. By explicitly decomposing multimodal interactions via B-spline-parameterized edge functions, KAN transforms traditional "black-box" feature fusion into mathematically analyzable compositions of univariate functions. This functional decomposition mechanism not only visualizes multimodal decision pathways but also quantifies the dynamic contributions of different modalities during feature space transformations, offering a novel theoretical framework and technical pathway for resolving interpretability challenges in multimodal fusion. Unlike traditional methods limited to local explanations or abstract representations, KAN ensures mathematical verifiability across the entire pipeline from input features to final decisions, establishing a fine-grained interpretability system while maintaining model performance.

2.3 Feature Reduction Methods

In deep learning, feature reduction methods can be categorized into supervised (e.g., CNN/Transformer), unsupervised (e.g., autoencoders/DBN), and semi-supervised (e.g., pseudo-label pretraining) [15]. Traditional supervised methods rely on implicit backpropagation for feature selection, lacking theoretical frameworks for noise separation. Unsupervised methods tend to retain task-irrelevant information due to the absence of label guidance. Semi-supervised methods suffer from cumulative pseudo-label noise. Although the Information Bottleneck (IB) theory provides an information-theoretic foundation for feature compression, its application in deep learning often faces limitations such as insufficient multimodal noise suppression and high computational complexity. The proposed DRD-MIB method addresses these issues through targeted mutual information optimization, explicitly filtering single-modal noise during dimensionality reduction while introducing inter-modal mutual information constraints to prevent cross-modal conflicts.

2.4 Imbalanced Multimodal Learning

In our study, the imbalance problem in multimodal learning poses a critical challenge, as it leads models to overly rely on dominant modalities while neglecting others. To address this, Peng et al. [37]

proposed the OGM-GE algorithm, which dynamically adjusts gradient updates across modalities. Wu et al. [52] introduced a conditional learning velocity method to balance learning processes by capturing relative learning speeds among modalities. Guo et al. [8] developed a Classifier-Guided Gradient Modulation (CGGM) approach that employs modality-specific classifiers to evaluate contributions and dynamically modulate gradient magnitudes.

However, these methods exhibit limitations in specific scenarios. For instance, CGGM [8] requires additional classifiers for gradient modulation and mandates fully connected layers for both single-modal and multi-modal classifiers, making it incompatible with our KAN-based architecture. The conditional learning velocity method by Wu et al. [52], designed for intermediate fusion strategies, struggles to scale to scenarios with more than two modalities. OGM-GE [37] focuses solely on gradient magnitude adjustment while ignoring directional alignment.

To overcome these issues, Wei et al. [47] proposed the MMPareto algorithm, which leverages multi-objective Pareto optimization to balance both gradient direction and magnitude during updates. MMPareto eliminates the need for auxiliary classifiers or specific architectures and supports arbitrary numbers of modalities, aligning well with our requirements. However, it fails to address the impact of intra-modal noise on Pareto frontiers. Our proposed MCPareto adopts a hierarchical optimization framework comprising two layers: (1) a feature purification layer that employs DRD-MIB to suppress single-modal noise and compress dimensions, filtering out emotion-unrelated interference; and (2) a gradient coordination layer that synergizes denoised features with MMPareto to significantly enhance the utilization of minor modalities (e.g., audio and visual), thereby improving overall model performance.

3 Model Architecture

In this section, we present the detailed architecture of KAN-MCP, a novel framework that integrates the interpretability of KAN with the robustness of the MCPareto framework. As shown in Figure 2, the proposed model addresses the challenges of interpretability and multimodal learning imbalance in MSA through a two-stage design: a feature encoding stage and a fusion-prediction stage. Below, we elaborate on each component and their synergistic integration.

3.1 Encoding Stage

The encoding stage is responsible for extracting and purifying features from linguistic, acoustic, and visual modalities. This stage addresses the issue of modality noise interference through the Dimensionality Reduction and DRD-MIB method, which is based on the IB) principle. DRD-MIB ensures that only the most relevant information is retained while filtering out noise, thereby providing clean and compact representations for downstream tasks.

3.1.1 Unimodal Feature Extraction. We first perform feature extraction on each modality separately to obtain latent representations. For the textual modality, we leverage pre-trained language models (e.g., DeBERTa [11] or BERT [4]) to tokenize and encode textual input I_t into a feature sequence $x_t \in \mathbb{R}^{d_t \times T_t}$. For the acoustic and visual modalities, we use preprocessed feature sequences $x_a \in \mathbb{R}^{d_a \times T_a}$ and $x_v \in \mathbb{R}^{d_v \times T_v}$, respectively. Temporal averaging is then applied to derive modality-specific representations.

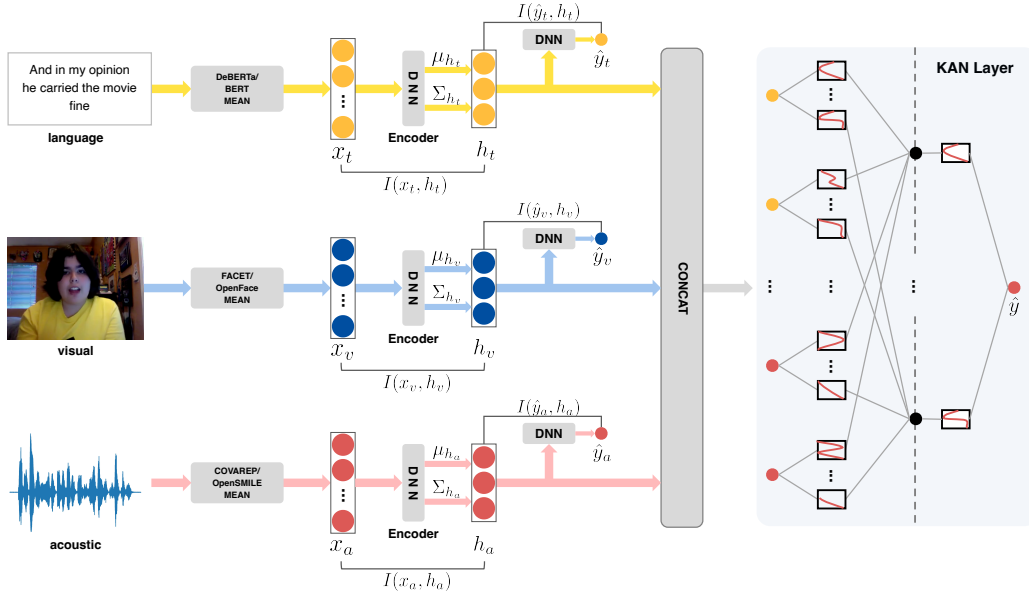


Figure 2: The overall architecture of the proposed KAN-MCP.

3.1.2 Unimodal Feature Reduction. To filter out noise and redundancy while achieving feature dimensionality reduction, we apply DRD-MIB to encode x_m into a minimally sufficient predictive representation h_m . DRD-MIB enables unimodal representations to filter out noisy information prior to fusion and aligns the encoded unimodal distributions by individually applying the Information Bottleneck (IB) principle to each modality. The objective of DRD-MIB is formulated as:

$$\mathcal{L}_{DRD-MIB} = I(y; h) + \sum_{m \in \{t, a, v\}} [I(y; h^m) - \gamma I(x^m; h^m)] \quad (1)$$

where y is the label, h is the fused multimodal representation, x^m denotes the unimodal representation output by the unimodal learning network, and h^m is the encoded unimodal representation.

DRD-MIB can be approximated by the following objective function. For brevity, the detailed derivation is relegated to the Supplementary Material, and we present only the final result here:

$$\begin{aligned} \mathcal{J}_{DRD-MIB} &\approx \frac{1}{b} \sum_{i=1}^b \left[\log Q(y_i | h_i) + \right. \\ &\quad \left. \sum_{m \in \{t, a, v\}} \left[\mathbb{E}_{\epsilon \sim \mathbb{P}(\epsilon)} [\log Q(y_i | h_i^m)] - \gamma \cdot KL \left(\mathbb{P}(h_i^m | x_i^m) \parallel q(h_i^m) \right) \right] \right] \\ &= \frac{1}{b} \sum_{i=1}^b \left[\log Q(y_i | h_i) + \sum_{m \in \{t, a, v\}} \left[\mathbb{E}_{\epsilon \sim \mathbb{P}(\epsilon)} [\log Q(y_i | h_i^m)] \right. \right. \\ &\quad \left. \left. - \beta \cdot KL \left(\mathcal{N}(\mu_{h_i^m}, \Sigma_{h_i^m}) \parallel \mathcal{N}(0, I) \right) \right] \right] \end{aligned} \quad (2)$$

where,

$$\begin{aligned} \mathbb{P}(h^m | x^m) &= \mathcal{N}(\mu^m(x^m; \theta_\mu^m), \Sigma(x^m; \theta_\Sigma^m)) \\ &= \mathcal{N}(\mu_{h^m}, \Sigma_{h^m}) \end{aligned} \quad (3)$$

$$h^m = \mu_{h^m} + \Sigma_{h^m} \times \epsilon \quad (4)$$

where b is the sampling size (i.e., batch size), and i is the subscript that indicates each sample, μ^m and Σ^m are the deep neural networks that learn the mean and the variance of $\mathbb{P}(h^m | x^m)$ respectively. From the perspective of filtering out noisy information before fusion, DRD-MIB is effective as it forces the encoded unimodal representation h^m to only contain the relevant information to the target. Moreover, by using the reparameterization trick, the distributions of different encoded unimodal representations are closer, which narrows down the modality gap.

3.2 Fusion-Prediction Stage

The fusion-prediction stage leverages KAN to achieve interpretable and efficient multimodal fusion. Unlike traditional neural networks, KAN places activation functions on the edges (connections) rather than nodes, enabling it to approximate complex multivariate functions through compositions of univariate functions. This design enhances interpretability by allowing direct analysis of feature transformations without external interpretation tools. Specifically, we process the compressed unimodal features h_t, h_a, h_v and concatenate them into a fused vector F , enabling efficient and interpretable representation for downstream tasks. Subsequently, KAN processes this fused representation to generate the final prediction:

$$\hat{y} = \text{KAN}(F) \quad (5)$$

To better understand how KAN achieves this, we introduce its unique architecture, which is inspired by the Kolmogorov-Arnold Representation Theorem. Unlike traditional multi-layer perceptrons (MLPs), KAN employs learnable activation functions on edges (weights rather) than fixed activations on nodes (neurons). This design not only enhances interpretability but also achieves higher accuracy with fewer parameters. Formally, KAN is defined as:

$$\text{KAN}(x) = (\Phi_{L-1} \circ \Phi_{L-2} \circ \dots \circ \Phi_1 \circ \Phi_0)(x) \quad (6)$$

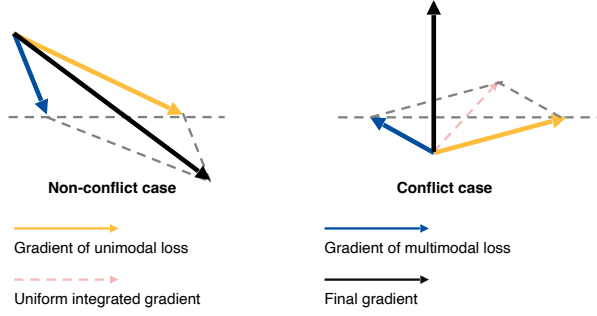


Figure 3: Gradient integration strategy of MMPareto.

where Φ_i denotes the i -th layer. Each layer with n_{in} inputs and n_{out} outputs is represented as a matrix of univariate functions:

$$\Phi = \{\phi_{q,p}\}, \quad p = 1, 2, \dots, n_{in}, \quad q = 1, 2, \dots, n_{out} \quad (7)$$

Here, $\phi_{q,p}$ is a univariate function with trainable parameters. The transformation from layer l to $l+1$ is expressed as:

$$\mathbf{x}_{l+1} = \begin{pmatrix} \phi_{l,1,1}(\cdot) & \phi_{l,1,2}(\cdot) & \cdots & \phi_{l,1,n_l}(\cdot) \\ \phi_{l,2,1}(\cdot) & \phi_{l,2,2}(\cdot) & \cdots & \phi_{l,2,n_l}(\cdot) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_{l,n_{l+1},1}(\cdot) & \phi_{l,n_{l+1},2}(\cdot) & \cdots & \phi_{l,n_{l+1},n_l}(\cdot) \end{pmatrix} \mathbf{x}_l, \quad (8)$$

This structure allows KAN to approximate complex functions via univariate function compositions, enhancing multimodal fusion and prediction performance. By employing edge-based activation functions, KAN achieves interpretability and efficiency, overcoming traditional neural networks' limitations.

3.3 Multimodal Learning Imbalance

This study proposes the MCPareto framework to address modality imbalance and noise interference in multimodal learning. To overcome the limitations of MMPareto algorithm that neglect noise in non-dominant modalities leading to performance degradation, we adopt a two-stage optimization approach: First, the DRD-MIB algorithm reduces single-modality noise while enhancing reliable gradients from non-dominant modalities. Subsequently, an MM-Pareto optimization algorithm coordinates gradient balance between single-modality and multimodal objectives. This dual-stage collaborative mechanism effectively enhances model robustness and comprehensive performance, overcoming the constraints of existing methods. For our task of simultaneously optimizing single-modality and multimodal gradients, the Pareto algorithm can be formulated as follows:

$$\begin{aligned} \min_{\alpha^m, \alpha^u \in \mathbb{R}} \quad & \|\alpha^m \mathbf{g}_S^m + \alpha^u \mathbf{g}_S^u\|^2 \\ \text{s.t.} \quad & \alpha^m, \alpha^u \geq 0, \\ & \alpha^m + \alpha^u = 1, \end{aligned} \quad (9)$$

where $\mathbf{g}_S^m$ and $\mathbf{g}_S^u$ denote the gradients of the multimodal loss and the unimodal loss over a minibatch, respectively. By computing the cosine similarity $\cos \beta$ between $\mathbf{g}_S^m$ and $\mathbf{g}_S^u$, potential conflicts can be detected. The MMPareto algorithm is illustrated in Figure 3.

Under non-conflicting conditions, the directions of gradient vectors are aligned, enabling their contributions to jointly benefit all learning objectives. This implies that during optimization, these gradients can simultaneously improve multiple objectives without mutual interference. In such cases, the weights α^m and α^u are both set to 0.5 (instead of using the Pareto analytical solution) to amplify the SGD noise term, thereby enhancing generalization capability. The final gradient is formulated as:

$$\mathbf{h}_S^{\text{MMPareto}} \sim \mathcal{N} \left(\mathbf{g}_N^m + \mathbf{g}_N^u, \frac{C^m + C^u}{|S|} \right) \quad (10)$$

Under conflicting conditions, the optimal weights α^m and α^u are derived by solving the Pareto optimization problem to identify a conflict-free direction. The parameter γ is then applied to amplify the SGD noise term. The final gradient is:

$$\mathbf{h}_S^{\text{MMPareto}} \sim \mathcal{N} \left(\lambda(\mathbf{g}_N^m + \mathbf{g}_N^u), \lambda^2 \frac{(2\alpha^m)^2 C^m + (2\alpha^u)^2 C^u}{|S|} \right) \quad (11)$$

where,

$$\lambda = \frac{\|\mathbf{g}_S^m + \mathbf{g}_S^u\|}{\|2\alpha^m \mathbf{g}_S^m + 2\alpha^u \mathbf{g}_S^u\|} > 1 \quad (12)$$

By integrating the MCPareto framework, our model architecture effectively mitigates imbalance in multimodal learning, ensuring that unimodal auxiliary tasks positively contribute to the primary multimodal learning objective.

4 Experiments

We conducted extensive experiments to evaluate KAN-MCP using the CMU-MOSI [58], CMU-MOSEI [59], and CH-SIMS v2 [21] datasets. Due to space constraints, the descriptions of the datasets, feature extraction details, evaluation metrics, training details, and baselines are provided in the Supplementary Material.

4.1 Results on MSA

To evaluate the performance of KAN-MCP against competitive baselines, we employed two well-established MSA datasets. The results in Table 1 demonstrate that KAN-MCP exhibits superior performance, consistently outperforming baseline methods across most evaluation metrics on both the CMU-MOSI and CMU-MOSEI datasets. Specifically, on CMU-MOSI, KAN-MCP achieves a 0.6% improvement in 7-class accuracy (Acc7) over ITHP [53], which also employs DeBERTa [11] as its language network, along with 0.9% enhancements in binary accuracy (Acc2) and F1 score. Notably, KAN-MCP significantly surpasses existing methods in MAE. For CMU-MOSEI, KAN-MCP outperforms the state-of-the-art ITHP algorithm by 1.7%, 0.6%, and 0.5% in Acc7, Acc2, and F1 score, respectively. Furthermore, Table 2 presents the robustness validation results of KAN-MCP in cross-lingual scenarios. On the Chinese dataset CH-SIMS v2, the model maintains leading performance across all metrics, including 5-class accuracy (Acc5), 3-class accuracy (Acc3), Acc2, F1 score, MAE, and correlation (Corr).

Table 1: The comparison with baselines on the CMU-MOSI dataset and CMU-MOSEI dataset. Apart from ULMD, MFON, ConFEDE and UniMSE, the results of baselines are derived from our experiments. The best results are highlighted in bold, and the runner-up results are indicated with underlines.

Model	CMU-MOSI					CMU-MOSEI				
	Acc7↑	Acc2↑	F1↑	MAE↓	Corr↑	Acc7↑	Acc2↑	F1↑	MAE↓	Corr↑
Graph-MFN[59]	34.4	80.2	80.1	0.939	0.656	51.9	84.0	83.8	0.569	0.725
MFON[41]	33.3	80.0	80.1	0.948	0.664	50.8	83.4	83.4	0.580	0.722
MMIM[9]	45.0	85.1	85.0	0.738	0.781	53.1	85.1	85.0	0.547	0.752
HyCon[34]	46.6	85.2	85.1	0.741	0.779	52.8	85.4	85.6	0.554	0.751
UniMSE†[13]	<u>48.7</u>	86.9	86.4	0.691	0.809	<u>54.4</u>	<u>87.5</u>	<u>87.5</u>	0.523	0.773
ConFEDE†[54]	42.3	85.5	85.5	0.742	0.782	54.9	85.8	85.8	0.522	0.780
MGCL[33]	49.3	86.7	86.7	0.685	0.707	53.9	86.4	86.4	0.535	0.772
ULMD†[64]	47.8	85.8	85.7	0.700	0.799	53.8	86.0	85.9	0.531	0.770
MFON† [63]	44.9	86.9	86.9	0.725	0.797	53.7	86.3	86.3	0.528	0.780
ITHP[53]	47.7	<u>88.5</u>	<u>88.5</u>	<u>0.663</u>	<u>0.856</u>	52.2	87.1	87.1	0.550	0.792
KAN-MCP	48.3	89.4	89.4	0.615	0.857	53.9	87.7	87.6	0.522	<u>0.788</u>

Table 2: The comparison with baselines on the CH-SIMS v2 dataset. The baseline models are reproduced with BERT as the language model. The best results are highlighted in bold, and the runner-up results are indicated with underlines.

Model	CH-SIMS v2					
	Acc5↑	Acc3↑	Acc2↑	F1↑	MAE↓	Corr↑
EF-LSTM [49]	53.7	73.5	80.1	80.0	0.309	0.700
LF-DNN [48]	51.8	71.2	77.8	77.9	0.322	0.668
TFN [56]	53.3	70.9	78.1	78.1	0.322	0.662
LMF [22]	51.6	70.0	77.8	77.8	0.327	0.651
MFN [57]	<u>55.4</u>	72.7	79.4	79.4	0.301	0.712
Graph-MFN [59]	48.9	68.6	76.6	76.6	0.334	0.644
MuT [43]	54.6	74.2	<u>80.8</u>	<u>80.7</u>	0.300	0.738
MISA [10]	47.5	68.9	78.2	78.3	0.342	0.671
MAG-BERT [38]	49.2	70.6	77.1	77.1	0.346	0.641
Self-MM [55]	53.5	72.7	78.7	78.6	0.315	0.691
MMIM [9]	50.5	70.4	77.8	77.8	0.339	0.641
AV-MC [21]	52.1	73.2	80.6	<u>80.7</u>	0.301	0.721
KuDA [6]	53.1	<u>74.3</u>	80.2	80.1	<u>0.289</u>	<u>0.741</u>
KAN-MCP	57.3	75.0	81.6	81.7	0.281	0.742

4.2 Study of Interpretability

In this section, we use the visualization capabilities of the KAN model to analyze the multimodal fusion prediction process, with results illustrated in Figure 4. By observing connection transparency, we can assess the contribution of different modalities. For the CMU-MOSI and CMU-MOSEI datasets, Figures 4(a) and 4(b) reveal that the text modality exhibits significantly lower connection transparency between data points and intermediate nodes compared to other modalities. This observation indicates that textual information plays a fundamental and critical role in the model, which aligns with existing research conclusions [30, 31] and further validates the pivotal status of textual information in sentiment analysis tasks.

In contrast, the audio and visual modalities show higher connection transparency. This suggests that audio and visual information are less directly impactful. However, this does not mean that the roles of the audio and visual modalities can be neglected. On the

contrary, they primarily serve indispensable supplementary functions during model operation. In specific scenarios, audio and visual information provide cues to assist the model. For instance, in complex multimodal emotional scenarios, textual information alone might fail to capture all affective clues, whereas audio and visual signals could offer multi-dimensional supplementary emotional hints to enhance comprehensive understanding of emotional expressions. This complementary effect not only enriches the model’s feature inputs but also optimizes its performance, enabling more flexible and accurate sentiment analysis and prediction in diverse real-world applications.

For the CH-SIMS v2 dataset, Figure 4(c) demonstrates relatively more balanced connection intensities between textual, audio, and visual data points and intermediate nodes, without a clearly dominant modality. This phenomenon stems from the comparable sentiment analysis capabilities of all three modalities in the CH-SIMS v2 dataset, a finding that is further corroborated in subsequent analyses of imbalanced multimodal learning.

Furthermore, KAN can provide more intuitive mathematical expressions and enhanced interpretability through symbolic manipulation. Related experimental results are available in the Supplementary Material.

4.3 Imbalanced Multimodal Learning Analysis

In this section, we present the learning curves of unimodal losses across three datasets. As illustrated in Figure 5, the training losses of the textual modality demonstrate a continuous decline across all three datasets, indicating the model’s pronounced advantages in textual feature extraction and sentiment semantic comprehension. Notably, the audio and visual modalities in CMU-MOSI and CMU-MOSEI datasets exhibit significant training instability, characterized by marked fluctuations in their loss curves. This phenomenon aligns with the imbalanced modality contribution patterns revealed in Figure 4. In contrast, the multimodal training process of CH-SIMS V2 dataset displays superior synergy, with non-textual modalities (audio, visual) demonstrating smooth descending loss curves whose convergence rates remain synchronized with the textual modality. These disparities in training stability effectively

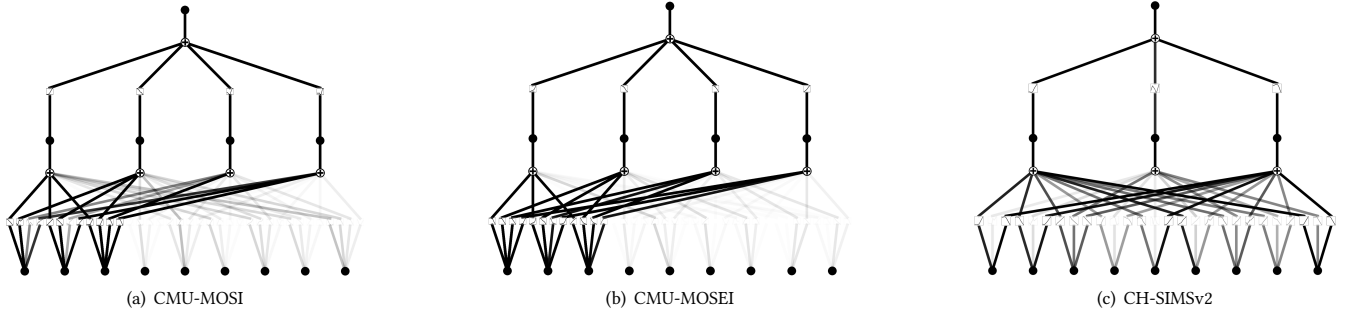


Figure 4: KAN Visualization Across Datasets: Opacity Negatively Correlates with Modality Contribution Weights.

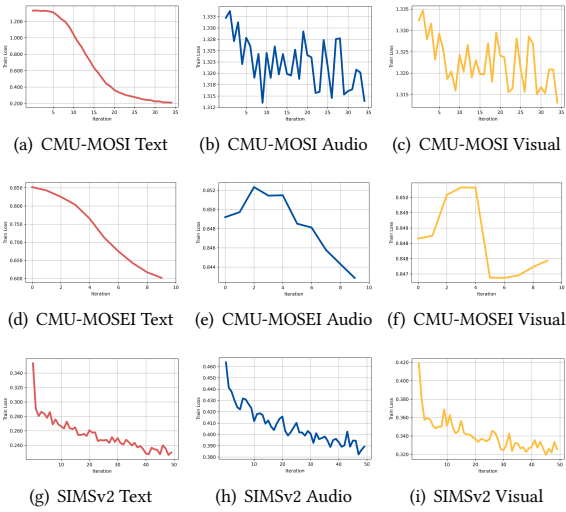


Figure 5: The Learning Curves of the Training Losses for Audio, Visual, and Text Modalities.

elucidate the observed differences in modality contributions across datasets for MSA tasks: When non-textual modalities achieve sufficient learning effectiveness (as in CH-SIMS V2), the model attains more balanced multimodal representation fusion; conversely, it may develop over-reliance on dominant modalities (e.g., textual modality in CMU datasets) when such effectiveness is lacking. This finding corroborates the aforementioned characteristics of modality contribution equilibrium.

The scarcity of semantic information in non-textual modalities within the CMU-MOSI and CMU-MOSEI datasets may stem from two primary factors. First, acoustic signals and visual information inherently contain noise interference, and the instability in raw data quality directly impacts the robustness of the feature extraction process. Second, the feature extraction architectures employed in these datasets (e.g., Facet and COVAREP) may inadequately capture discriminative features pertinent to sentiment analysis tasks.

From the above analysis, it is evident that the quality of modal representations exhibits a significant correlation with the transparency of connecting lines in KAN visualization diagrams. By

Table 3: Comparison and Performance Analysis of DRD-MIB and Common Dimensionality Reduction Methods (Autoencoder, Transformer, Fully Connected Layer) on the CMU-MOSEI Dataset Under MMPareto Conditions.

	MMPareto	Acc7	Acc2	F1	MAE	Corr
FC	w/o	53.2	86.4	86.3	0.527	0.777
	w	52.0	85.4	85.4	0.557	0.753
AE	w/o	53.0	86.4	86.4	0.532	0.779
	w	47.3	82.6	82.2	0.614	0.703
Transformer	w/o	53.0	86.4	86.3	0.542	0.779
	w	50.9	85.0	85.0	0.570	0.768
DRD-MIB	w/o	53.1	86.5	86.4	0.536	0.782
	w	53.9	87.7	87.6	0.522	0.788

Table 4: Ablation experiments on CMU-MOSEI dataset.

	Acc7(↑)	Acc2(↑)	F1(↑)	MAE(↓)	Corr(↑)
w/o DRD-MIB	47.3	82.6	82.2	0.614	0.703
w/o KAN	52.9	86.7	86.6	0.535	0.780
w/o MMPareto	53.1	86.5	86.4	0.536	0.782
KAN-MCP	53.9	87.7	87.6	0.522	0.788

observing the transparency of connection lines between modal data points and intermediate nodes, the quality of modal representations can be indirectly assessed. For instance, high-quality modalities typically manifest as connection lines with lower transparency (more opaque), whereas low-quality modalities are characterized by lines with higher transparency (more translucent). This visual analytical approach provides an intuitive tool for understanding the relationship between modal representation quality and model dependency, while also offering valuable insights for optimizing the design of multimodal models.

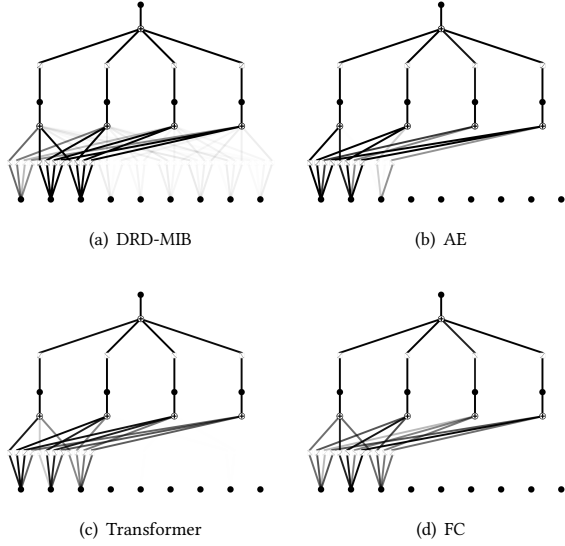
4.4 Ablation Study

In this section, we conducted extensive ablation studies to evaluate the effectiveness of each component in KAN-MCP:

4.4.1 The importance of KAN. In the "W/O KAN" setting, we replaced KAN with a multilayer perceptron (MLP). As shown in Table 4, removing KAN caused a 0.4% drop in Acc7, a more significant 1.2% decrease in Acc2, and a 1.1% reduction in F1 score.

Table 5: Comparative Analysis of MMPareto’s Effectiveness.

Dataset	MMPareto	Acc2	F1	MAE	Corr
CMU-MOSI	w/o	87.3	87.3	0.647	0.839
	w	89.4	89.4	0.615	0.857
CMU-MOSEI	w/o	86.5	86.4	0.536	0.782
	w	87.7	87.6	0.522	0.788
CH-SIMSV2	w/o	81.2	81.3	0.282	0.743
	w	81.6	81.7	0.281	0.742

**Figure 6: KAN Visualization of DRD-MIB and Common Dimensionality Reduction Methods (AutoEncoder, Transformer, Fully Connected Layer) on the CMU-MOSEI Dataset Under MMPareto-Free Conditions.**

These results confirm that KAN enhances interpretability while significantly improving accuracy in multimodal fusion prediction.

4.4.2 Discussion on DRD-MIB. In the "W/O DRD-MIB" setting, as shown in Table 4, the performance of KAN-MCP exhibited significant degradation when an autoencoder (AE) was used as the feature dimensionality reduction component. To further investigate this phenomenon, we removed the MMPareto algorithm and compared DRD-MIB with conventional dimensionality reduction methods, including AE, Transformer, and fully connected layers (FC). The results in Table 3 indicate that, while conventional methods achieved comparable overall performance to DRD-MIB, an in-depth analysis of KAN’s visualization results, as shown in Figure 6(b), 6(c), and 6(d), revealed a critical limitation: traditional methods predominantly captured textual modality information while neglecting audio and visual modality features. This highlights the inherent inability of conventional dimensionality reduction methods to effectively integrate multimodal information, particularly in scenarios where certain modalities contain limited information content.

In contrast, when DRD-MIB was employed as the dimensionality reduction component, as shown in Figure 6(a), the model successfully learned textual modality features while simultaneously capturing audio and visual modality information, thereby achieving comprehensive and effective multimodal integration. These results further validate the superiority of DRD-MIB in feature dimensionality reduction. Specifically, DRD-MIB leverages its unique mechanism to learn more balanced and high-quality unimodal representations, thereby significantly enhancing model performance in multimodal tasks. This finding underscores the distinct advantages of DRD-MIB in multimodal feature dimensionality reduction and information integration.

4.4.3 Discussion on MMPareto. In the "W/O MMPareto" setting, we analyzed performance across three datasets. As shown in Table 5, performance declined across all datasets, with notable degradation observed on CMU-MOSI and CMU-MOSEI, where Acc2 and F1 scores decreased by 2.1% and 1.2%, respectively. In contrast, CH-SIMS v2 exhibited only marginal declines (0.4% in Acc2 and F1). These results suggest that MMPareto effectively leverages auxiliary unimodal information to enhance multimodal learning performance when significant inter-modal information disparities exist and weaker modalities contain non-harmful yet useful information. However, its performance gains diminish when modalities exhibit relatively balanced information content.

Further comparative experiments revealed compatibility issues between traditional dimensionality reduction methods (AE, Transformer, FC) and the MMPareto optimizer. As shown in Table 3, unexpected performance degradation occurred when conventional dimensionality reduction components were used. We hypothesize that this stems from the inability of traditional methods to effectively filter redundant features, leading to increased gradient conflicts between unimodal and multimodal loss directions, thereby impairing model performance. This suggests that the efficacy of MMPareto may be constrained when weak modality features suffer from information loss.

In summary, our proposed MCPareto framework integrates the DRD-MIB denoising dimensionality reduction method and the MMPareto optimization algorithm. This combination suppresses noise interference while enhancing the contribution weights of secondary modalities, thereby improving model generalization. This two-stage optimization strategy not only addresses modality imbalance but also significantly enhances robustness in complex multimodal scenarios, offering a more reliable solution for multimodal learning.

5 Conclusion

This paper proposes a novel MSA framework, KAN-MCP, designed to address the interpretability of multimodal fusion and modality imbalance issues. By integrating KAN with MCPareto, the framework achieves transparent cross-modal interaction modeling and balanced modality learning. Experimental results demonstrate that KAN-MCP significantly outperforms existing methods on CMU-MOSI, CMU-MOSEI, and CH-SIMS v2 datasets. Furthermore, KAN’s visualization module intuitively reveals the contributions of different modalities in sentiment judgment, validating the dominance of textual modality in CMU datasets and the synergistic balance effect among modalities in the CH-SIMS v2 dataset.

Acknowledgments

This work was supported by a grant from South China Normal University (Grant No. 24KJ01).

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